

Statement:

A practical guide of evaluation and management of HIV infected children

Introduction:

Children with HIV are special group of pediatric patients that needs close follow-up and expert management. The complexity of their care is related to the nature of the infection in addition to the social back ground and challenges they face in the community. In Oman, Royal hospital is the only tertiary hospital has specialized clinic for HIV exposed and infected children. It is a weekly clinic (Monday and Thursday) where infected children till age of 13 years are seen and followed-up by the PID team. This document is made to make a practical guideline and standardized approach to diagnosis and management of both the exposed and infected children with HIV. It is also going to be a practical guideline for trainee joining team and handy tool for planning their patients care.

HIV exposure in children:

- Most of the children followed in the clinic are those with antenatal exposure to HIV in the mother during pregnancy and HIV infection. Few cases get referred as exposure to breast milk, needle prick, blood transfusion related or sexual assault.
- Most HIV infections in children are acquired via mother-to-child transmission during pregnancy, labor, delivery, or breastfeeding.
- Prevention of mother to child transmission (PMTCT) program started on 2009 with introduction of HIV screening to all pregnant ladies attending antenatal care clinics with an acceptance rate of 99 percent.

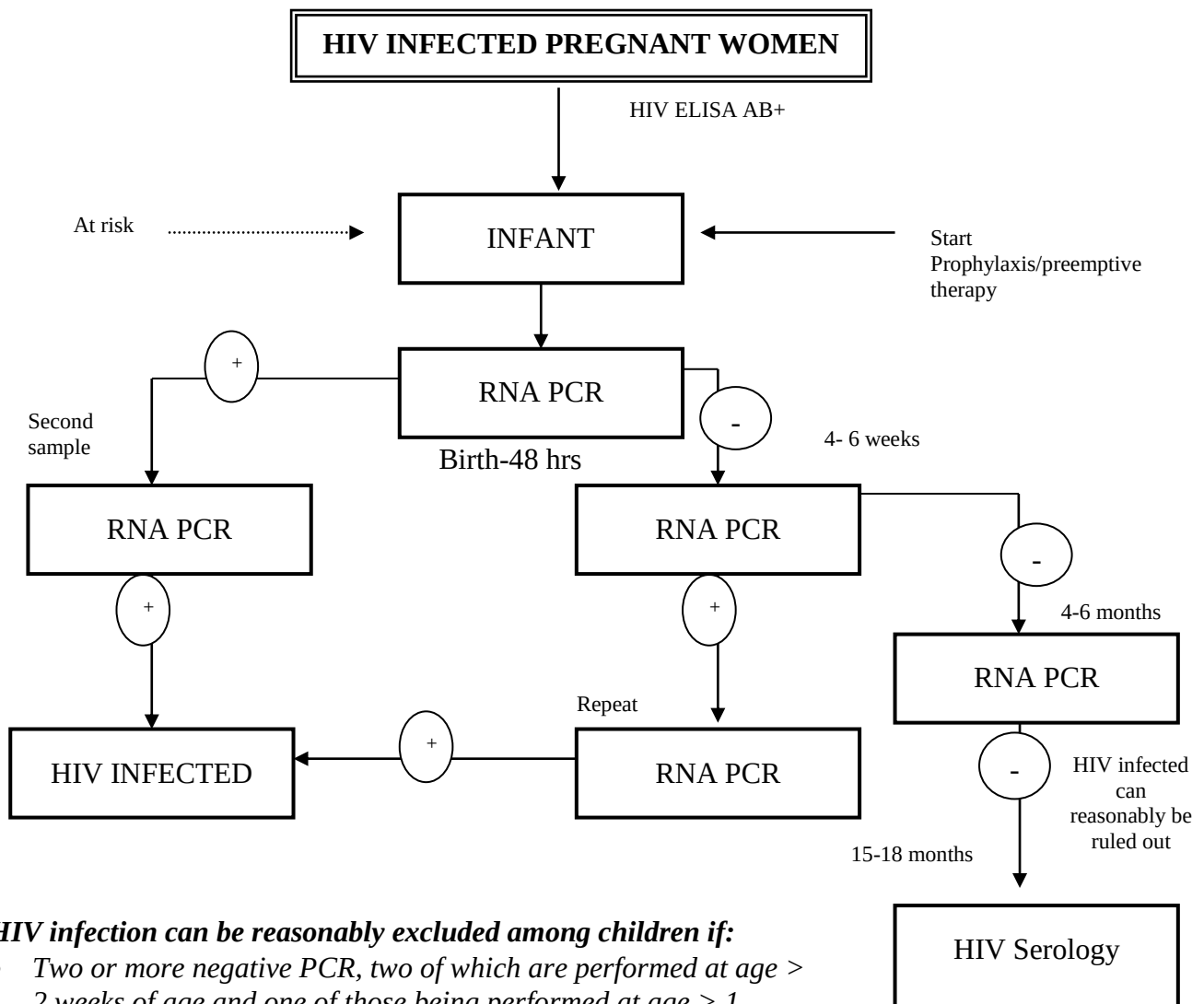
Testing for HIV in pregnant women:

- HIV serology is recommended for all pregnant women in each pregnancy irrespective of prior HIV testing.
- Un-booked women coming for delivery or un-booked home deliveries coming in postnatal period will also be tested for HIV.
- For positive HIV, women's partner, children below the age of 15 years and other contacts will be tested for HIV.

Testing for HIV infection in infants born to mothers with HIV infection:

- 1- newborn samples HIV RNA PCR and CBC within 48 hours of birth
- 2- Second HIV RNA PCR at 4- 6 weeks of age, and a 3rd test should be between 4-6 months of age for definite exclusion.
- 3- HIV serology at 18 months of age (Figure 1)

Figure 1: Management of antenatal exposure of HIV



HIV infection can be reasonably excluded among children if:

- Two or more negative PCR, two of which are performed at age > 2 weeks of age and one of those being performed at age ≥ 1 months.

HIV can definitely be excluded if:

- Two or more negative PCR, two at which are performed at age > 1 month and one of those being performed at age ≥ 4 months.
OR
One test of negative HIV antibodies at > 6 months

Cord blood should not be used for testing

- The newborn risk of perinatal transmission should be assessed to decide the regimen of therapy and duration. **Table 1**
- Zidovudine, at gestational age-appropriate doses, should be initiated as close to the time of birth as possible, preferably within 6 to 12 hours of delivery. **Table 2**

- The ID team will see the mother and baby prior to discharge to insure appropriate investigation and management plan was started and for counseling of the parents and arrangement of follow-up.
- BCG vaccine should not be given to infants born for HIV positive mothers and a note should be put in vaccine card to not give the baby any life attenuated vaccine. Please cross over OPV in vaccine card and put IPV instead.

**** NO NEED TO WRITE IN THE CARD THAT BABY IS EXPOSED TO HIV AND PLEASE MAINTAIN CONFIDENTIALITY OF THE INFORMATION.**

- All exposed babies will be followed in PID clinic at age of 4- 6 weeks, 4-6 months and at 18 months for follow up and further management.
- To prevent Pneumocystis jirovecii pneumonia (PCP), all infants born to women with HIV infection should begin PCP prophylaxis (cotrimoxazole), check dose from ARV drug doses below) at ages 4 to 6 weeks, after completing their ARV prophylaxis regimen, unless there is adequate test information to presumptively exclude HIV infection. The exposed infant will remain on PCP prophylaxis until HIV excluded (see Figure 1).
- After rolling out HIV infection in newborn the BCG vaccine will be given and infant can have other life attenuated vaccines.

**** HIV infected mothers can also have other STD so please check mother Hepatitis status (HBV, HCV) and syphilis. The maternal status of those infections will determine need for testing, prophylaxis and management of their newborn.**

Counseling HIV positive mother after delivery:

- Before starting counseling make sure of the following information:
- Detailed history on the pregnancy
- Viral load during and before delivery
- Maternal viral resistance genotyping
- Medication received during pregnancy and compliance
- Medication during delivery
- Father status and his viral load if it is available
- Presence or exclusion of other STD in both parents.

Counseling mother with controlled viral load

- Counseling should be started once mother is free of pain especially if she delivered by C- section
- It should be done in a side room and confidentiality need to be taken
- Congratulate both parents of the new baby
- Clarify parental understanding
- Make sure if it is first baby or they have other kids and if they are tested and their status

- Explain HIV test that is already or going to be done to the newborn and how frequent should be repeated
- Explain risk of transmission
- Explain importance of the prophylaxis
- Explain avoiding breast feeding and vaccination management in the newborn
- Check understanding
- Give follow up

Counseling mother with uncontrolled viral load

- Follow the above points
- Explain that there is risk of transmission of the infection with uncontrolled viral load
- Explain to avoid breast feed
- Explain importance of prophylaxis and close follow up of tests

Counseling HIV negative mother with HIV positive partner

- If mother have been tested soon before delivery and she is negative, so there is no risk of transmission during delivery.
- Breast feeding is still contraindicated because mother is in continuous exposure to her husband (especially if father viral load is not controlled).

Table 1

Level of perinatal HIV transmission risk	Description	Neonatal ARV management

Low Risk of Perinatal HIV Transmission	Infants ≥ 37 weeks gestation when the mother— - Is currently receiving and has received at least 10 consecutive weeks of ART during pregnancy, <i>and</i> - Has achieved and maintained <i>or</i> maintained viral suppression (defined as at least two consecutive tests with HIV RNA levels < 50 copies/mL obtained at least 4 weeks apart) for the remainder of the pregnancy, <i>and</i> - Has HIV RNA < 50 copies/mL at or after 36 weeks and within 4	ZDV for 2 weeks ^a
	- weeks of delivery, <i>and</i> - Did not have acute HIV infection during pregnancy, <i>and</i> - Has reported good ART adherence, and adherence concerns have not been identified.	

	Infants born to mothers who do not meet the criteria above or criteria for high risk below but who have a HIV RNA <50 copies/mL at or after 36 weeks gestation	ZDV for 4 to 6 weeks ^a
	Premature infants (<37 weeks gestation) who are not at high risk of perinatal acquisition of HIV	ZDV for 4 to 6 weeks ^a
High Risk of Perinatal HIV Transmission^{a,b}	Mothers who did not receive antepartum ARV drugs, <i>or</i> Mothers who received only intrapartum ARV drugs, <i>or</i> Mothers who received antepartum ARV drugs but did not have viral suppression (defined as at least two consecutive tests with HIV RNA level <50 copies/mL obtained at least 4 weeks apart) within 4 weeks prior to delivery, <i>or</i> Mothers with acute or primary HIV infection during pregnancy or breastfeeding (in which case, breastfeeding should be immediately discontinued)^c	Presumptive HIV therapy using either ZDV, 3TC, and NVP (treatment dose) refer to table 2, administered together from birth for 6 weeks

Presumed Newborn HIV Exposure	Mothers with unconfirmed HIV status who have at least one positive HIV test at delivery or postpartum, <i>or</i> Mothers whose newborn has a positive HIV antibody test	ARV management as described above for newborns with a high risk of perinatal HIV acquisition Infant ARV drugs should be discontinued immediately if supplemental testing confirms that the mother does not have HIV.
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3TC = lamivudine; ART = antiretroviral therapy; ARV =antiretroviral; NVP = nevirapine; ZDV = zidovudine

Table 1: Risk of perinatal HIV transmission and regimen of therapy

All HIV exposed infants (initiated as soon after delivery as possible)		
Drug	Age	Doses
ZDV	≥35 Weeks Gestation at Birth	<i>Birth to Age 4 Weeks</i> • ZDV 4 mg/kg per dose orally twice daily or alternative simplified weight-band dosing (see below) <i>Age >4 Weeks</i> • ZDV 12 mg/kg per dose orally twice daily; only make this dose increase for infants with confirmed HIV infection.
	≥30 to <35 Weeks' Gestation at Birth	<i>Birth to Age 2 Weeks</i> • ZDV 2 mg/kg per dose orally twice daily <i>Age 2 Weeks to 6 to 8 Weeks</i> • ZDV 3 mg/kg per dose orally

		twice daily <i>Age >6 to 8 Weeks</i> • ZDV 12 mg/kg per dose orally twice daily; make this dose increase only for infants with confirmed HIV infection.
ZDV	<30 Weeks' Gestation at Birth	<i>Birth to Age 4 Weeks</i> • ZDV 2 mg/kg per dose orally twice daily <i>Age 4 to 8 to 10 Weeks</i> • ZDV 3 mg/kg per dose orally twice daily <i>Age >8 to 10 Weeks</i> • ZDV 12 mg/kg per dose orally twice daily; only make this dose increase for infants with confirmed HIV infection.
Additional antiretroviral prophylaxis agents for HIV exposed infants of women who received No antepartum antiretroviral prophylaxis (initiated as soon after delivery as possible) In addition to ZDV as shown above		

Nevirapine (NPV)	≥37 Weeks' Gestation at Birth	<i>Birth to Age 4 Weeks</i> - NVP 6 mg/kg per dose orally twice daily <i>Age >4 Weeks</i> - NVP 200 mg/m² BSA per dose orally twice daily; only make this dose increase for infants with confirmed HIV infection.
NPV	≥37 Weeks' Gestation at Birth	<i>Birth to Age 1 Week</i> - NVP 4 mg/kg per dose orally twice daily <i>Age 1 to 4 Weeks</i> - NVP 6 mg/kg per dose orally twice daily <i>Age >4 Weeks</i> - NVP 200 mg/m² BSA per dose orally twice daily; only make this dose increase for infants with confirmed HIV infection.
NPV	≥32 to <34 Weeks' Gestation at Birth	<i>Birth to Age 2 Weeks</i> - NVP 2 mg/kg per dose orally twice daily <i>Age 2 to 4 Weeks</i> - NVP 4 mg/kg per dose orally twice daily <i>Age 4 to 6 Weeks</i> - NVP 6 mg/kg per dose orally

		twice daily <i>Age >6 Weeks</i> • NVP 200 mg/m² BSA per dose orally twice daily; only make this dose increase for infants with confirmed HIV infection.
3TC	≥32 Weeks' Gestation at Birth	<i>Birth to Age 4 Weeks</i> • 3TC 2 mg/kg per dose orally twice daily <i>Age >4 Weeks</i> • 3TC 4 mg/kg per dose orally twice daily

3TC = lamivudine; ART = antiretroviral therapy; ARV = antiretroviral; NVP = nevirapine; ZDV = zidovudine

Table 2: Recommended neonatal doses for prevention of maternal-to-child HIV transmission

HIV disease:

For the diagnosis of HIV in children you need to consider the following:

- 1- Virologic assays(HIV PCR) that directly detect HIV must be used to diagnose HIV infection in infants younger than 18 months; antibody tests should not be used.
- 2- A positive virologic test should be confirmed as soon as possible by a repeat virologic test on a second specimen.
- 3- Diagnosis of HIV infection in children with non-perinatal exposure or children with perinatal exposure aged >24 months relies primarily on the use of HIV antibody tests.
- 4- When acute HIV infection is suspected, testing with an HIV PCR may be necessary to diagnose HIV infection.

HIV categories:

There are different ways to categories HIV:

- 1- Clinical categories
- 2- Immunological categories

Mild HIV-Related Symptoms

Children with two or more of the conditions listed but none of the conditions listed in Moderate Symptoms category

- Lymphadenopathy (≥ 0.5 cm at more than 2 sites; bilateral at 1 site)
- Hepatomegaly
- Splenomegaly
- Dermatitis
- Parotitis
- Recurrent or persistent upper respiratory tract infection, sinusitis, or otitis media

Moderate HIV-Related Symptoms

- Anemia (hemoglobin < 8 g/dL [< 80 g/L]), neutropenia (white blood cell count $< 1,000/\mu\text{L}$ [$< 1.0 \times 10^9/\text{L}$]), and/or thrombocytopenia (platelet count $< 100 \times 10^3/\mu\text{L}$ [$< 100 \times 10^9/\text{L}$]) persisting for ≥ 30 days
- Bacterial meningitis, pneumonia, or sepsis (single episode)
- Candidiasis, oropharyngeal (thrush), persisting (> 2 months) in children older than age 6 months
- Cardiomyopathy
- Cytomegalovirus infection, with onset before 1 month
- Diarrhea, recurrent or chronic
- Hepatitis
- Herpes simplex virus stomatitis, recurrent (> 2 episodes within 1 year)
- Herpes simplex virus bronchitis, pneumonitis, or esophagitis with onset before 1 month
- Herpes zoster (shingles) involving at least 2 distinct episodes or more than 1 dermatome
- Leiomyosarcoma
- Lymphoid interstitial pneumonia or pulmonary lymphoid hyperplasia complex
- Nephropathy
- Nocardiosis
- Persistent fever (lasting > 1 month)
- Toxoplasmosis, onset before 1 month
- Varicella, disseminated (complicated chickenpox)

Stage-3-Defining Opportunistic Illnesses In HIV Infection

- Bacterial infections, multiple or recurrent^a
- Candidiasis of bronchi, trachea, or lungs
- Candidiasis of esophagus
- Cervical cancer, invasive^b
- Coccidioidomycosis, disseminated or extrapulmonary
- Cryptococcosis, extrapulmonary
- Cryptosporidiosis, chronic intestinal (> 1 month duration)
- Cytomegalovirus disease (other than liver, spleen, or nodes), onset at age > 1 month
- Cytomegalovirus retinitis (with loss of vision)
- Encephalopathy attributed to HIV^c
- HSV: chronic ulcers (> 1 month duration) or bronchitis, pneumonitis, or esophagitis (onset at age > 1 month)
- Histoplasmosis, disseminated or extrapulmonary
- Isosporiasis, chronic intestinal (> 1 month duration)
- Kaposi sarcoma
- Lymphoma, Burkitt (or equivalent term)
- Lymphoma, immunoblastic (or equivalent term)
- Lymphoma, primary, of brain

Clinical categories of HIV in children below 13 years of age according to CDC classification**Immunological categories of children (CDC guidelines 2015)**

Stage	Age on Date of CD4 Test					
	<1 Year	%	1 to <6 Years	%	≥6 years	%
	Cells/μL		Cells/μL		Cells/μL	
1	≥1,500	≥34%	≥1,000	≥30	≥500	≥26
2	750–1499	26–33	500–999	22–29	200–499	14–25
3	<750	<26	<500	<22	<200	<14

^a The stage is based primarily on the CD4 count; the CD4 count takes precedence over the CD4 percentage, and the percentage is considered only if the count is missing. If a Stage 3-defining opportunistic illness has been diagnosed (Table 6), then the stage is 3 regardless of CD4 test results.

Opportunistic infections:

- HIV-related immunosuppression significantly increases the risk for acquiring opportunistic infections due to bacteria, viruses, fungi, and protozoa. It is considered as a major source of morbidity and mortality in HIV-infected patients.

The most common OIs in children are:

- Serious bacterial infections (most commonly pneumonia, often presumptively diagnosed, and bacteremia)
- Herpes zoster
- Disseminated Mycobacterium avium complex (MAC)
- Pneumocystis jirovecii pneumonia (PCP)
- Candidiasis (esophageal and tracheobronchial disease)
- Less commonly observed OIs (event rate <1 per 100 child-years) included CMV disease, cryptosporidiosis, TB, systemic fungal infections, and toxoplasmosis

Children appear to follow two basic patterns in the progression of untreated infection:

- 1- A rapid progression of disease in which children reach severe clinical and/or immunologic stages within the first year of life.
- 2 - or a more typical, slower progression, with clinical and immunologic deterioration by five to six years of life.

Chemoprophylaxis for PCP and MAC:

Indication for Cotrimoxazole prophylaxis in children:

- 1- HIV exposed infants at 4-6 weeks of age
- 2- HIV infected Infants less than 1 year of age regardless of CD4 cell count/percentage
- 3- HIV infected children aged 1 to <6 years with CD4 <500 cell/mm³ or < 15%
- 4- HIV infected children aged 6-12 years with CD4 < 200cells/mm³ or < 15%

Criteria for discontinuing Cotrimoxazole prophylaxis: note: do not discontinue in HIV infected children aged < 1 year

- 1- Age 1 to 6 years, CD4 $\geq 15\%$ or ≥ 500 cells/mm³ for > 3 consecutive months
- 2- Aged ≥ 6 years, CD4 $\geq 15\%$ or ≥ 200 cells/mm³ for > 3 consecutive months

Several dosing schemes are available:

- 1- Given 3 days per week on consecutive days or on alternate days , TMP 2.5-5mg/kg/dose twice daily
- 2- Given 2 days per week on consecutive days or on alternative days, TMP 2.5-5mg/kg/dose twice daily
- 3- Given every day as single daily dose of TMP 5-10 mg/kg

Cotrimoxazole common side effect:

- Agranulocytosis
- Skin rash
- Steven-Johnson syndrome
- Aplastic anaemia

Indication for MAC prophylaxis:

- 1- Aged < 1 year with CD4 count 750cells/mm³
- 2- Aged 1 to < 2 years with CD4 count < 500 cells/mm³
- 3- Aged 2 to < 6 years with CD4 count < 75 cells/mm³
- 4- Aged ≥ 6 years with CD4 count < 50 cells/mm³

Criteria for discontinuing MAC prophylaxis: note: do not discontinue in children < 2 years of age.

- 1- Aged 2 to < 6 years with CD4 > 200 cells/mm³ for > 3 consecutive months
- 2- Aged ≥ 6 years with CD4 > 100 cells/mm³ for > 3 consecutive months.

Choice: clarithromycin 7.5mg/kg/dose twice daily oral or
Azithromycin 20 mg/kg/dose once weekly oral

Immune Reconstitution Inflammatory Syndrome (IRIS):

- Clinical definition is new appearance or worsening of clinical illness temporally related to starting cART accompanied by ≥ 1 log₁₀ copies/ml decrease in plasma HIV RNA , that is not explained by newly acquired infection or disease, the usual course of a previously acquired disease, or cART toxicity with improvement in CD4 count.

Two types of IRIS:

- Paradoxical IRIS, reconstitution of the immune system produces a worsening of symptoms of an existing active OI .In Follow up cultures viable pathogens may be isolated
- Unmasking IRIS, the appearance of new symptoms of a latent or occult OI. Follow-up cultures most often are sterile.

In children, the conditions most commonly associated with IRIS include:

- Mycobacterial infections, herpes zoster, herpes simplex virus and cryptococcal infection.
- Reactions related to bacille Calmette-Guérin vaccine have been one of the most common IRIS manifestations in children in low-resource settings.

Risk factors for IRIS:

- Presence of opportunistic infection at the time of initiation of ART
- A low baseline CD4 cell count
- The time interval between the treatment of opportunistic infection and the initiation of ART, particularly in context of TB-IRIS (a shorter interval is associated with a higher risk of IRIS in these patients)

Hepatitis B or C	• Transient elevations in transaminases may occur after initiation of ARV therapy with immune reconstitution and can be difficult to distinguish from drug-induced hepatitis • Hepatic flares are usually mild and self-limited but can result in decompensation in someone with preexisting cirrhosis
Progressive multifocal Leukoencephalopathy	• PML lesions may be unmasked or worsen and could appear as new or worsening focal neurologic deficits or lesions on MRI scans
Kaposi's sarcoma	• Presents as worsening of Kaposi's sarcoma • Fatal IRIS has occurred in patients with preexisting Kaposi's sarcoma and multicentric Castleman disease after initiating ARV therapy • The frequency of human herpesvirus-8-associated IRIS is not known
Autoimmune diseases	• Preexisting autoimmune disorders such as sarcoidosis or Grave's disease may be exacerbated
Minor presentations	
Herpes simplex virus (HSV) and varicella zoster virus (VZV)	• HSV and VZV can reactivate after initiation of ARV therapy • Presentations are usually similar to non-IRIS disease; however, IRIS may worsen a patient's symptoms • Some patients become aware of their HSV infection only after the presentation of IRIS
Nonspecific dermatologic complications	• A number of dermatologic manifestations, such as folliculitis and oral and genital warts, may appear or worsen during immune reconstitution

Management of IRIS:

- For mild cases, observation alone with close clinical and laboratory monitoring may be sufficient.
- For moderate cases, nonsteroidal anti-inflammatory drugs have been used
- For severe cases, corticosteroids such as dexamethasone have been used. However, the optimal dose and duration of therapy are unknown, and inflammation can take weeks to months to subside.
- During that time, ART generally is continued.

First visit of HIV patient:

Entry into Care—Baseline Evaluation

At entry into care, HIV-infected children should have the following:

- A complete age-appropriate medical history

- Physical examination, and laboratory evaluation (see Table). This should include:
- General medical and social history (e.g., immunizations, nutrition, physical and social environment), evaluation for HIV-specific physical conditions (e.g., growth delay, microcephaly, motor or cognitive neurologic problems)
- Evaluation for HIV-associated laboratory abnormalities (e.g., anemia, leukopenia, thrombocytopenia, elevated glucose, transaminases or creatinine, hypoalbuminemia)
- Assessment of presence or risk of opportunistic infections (see Opportunistic Infections).
- Laboratory confirmation of HIV infection should be obtained if available documentation is incomplete (see HIV Diagnosis).
- CD4 cell count and percentage, as well as plasma HIV RNA measurements (i.e., viral load), should be obtained at entry into care to help guide decisions about timing of cART initiation.
- Genotype resistance testing should be performed, even if cART is not initiated immediately.
- Hepatitis B serology, CMV serology and toxoplasma serology
- If history or examination are suggestive of any of OI investigation should be guided according to.
- TST skin test

OPD visit:

The following checklist needs to be reviewed with each visit:

Monitoring patients with HIV in routine OPD visit

	Entry into care	pretherapy	CART initiation	Week 2-4 after therapy	Every 3-4 months	Only required every 6 months	ARV switch
History and physical	•	•	•	•	•	•	•
Adherence evaluation		•	•	•	•	•	•
CD4 count/percentage	•	•	•		• ^a	• ^b	•
plasma viral load	•	•	•		• ^a	• ^b	•
Resistance testing	•						•
CBC	•	•	•			•	•
Chemistry	•	•	•			•	•
Lipid profile	•		•			•	•
Urinalysis	•		•				•
Hepatitis B screen			•				•

^a if patient started on treatment within 1 month or more or uncontrolled viral load
^bCD4 cell count and percentage can be monitored less frequently (every 6 to 12 months) in children and youth who are adherent to therapy and have CD4 cell value well above the threshold for opportunistic infection risk, sustained viral suppression, and stable clinical status for more than 2 to 3 years

Management of infants and children with HIV:

Treatment recommendation:

All HIV infected children should be started on ART regardless of the age at diagnosis

Points need to considered:

- All children with HIV 12 months and below should be on treatment and cotrimoxazole prophylaxis despite their clinical and immunological category or viral load.

Antiretroviral drugs:

Standard recommendation regimens of ARV:

Preferred Initial Regimens Based on Age and Weight at Time of Treatment Initiation			
Age	Weight Restriction	Regimens	FDC Available (see Appendix A, Table 1)
Newborns, Birth to Age <14 Days ^{a,b}	None	Two NRTIs plus NVP	No
	≥2 kg	Two NRTIs plus RAL ^c	No
Neonates ≥14 Days to Age <4 Weeks	None	Two NRTIs plus LPV/r ^b	No
	≥2 kg	Two NRTIs plus RAL ^c	No
Infants and Children Aged ≥4 Weeks	≥3 kg	Two NRTIs plus DTG ^d	No
		Two NRTIs plus DTG ^d	Yes (≥10 kg)
Children Aged ≥2 Years	≥14 kg	Two NRTIs plus BIC ^e	Yes
Adolescents Aged ≥12 Years With SMRs of 4 or 5	Refer to the Adult and Adolescent Antiretroviral Guidelines .		Yes
Preferred Dual-NRTI Backbone Options for Use in Combination with Other Drugs			
Age	Dual-NRTI Backbone Options		FDC Available
Neonates Aged Birth to 1 Month	ABC plus (3TC or FTC) ^f		No ^g
	ZDV plus (3TC or FTC) ^h		No ^g
Infants and Children Aged >1 Month to <2 Years	ABC plus (3TC or FTC) ^f		Yes
	ABC plus (3TC or FTC) ^f		Yes

The preferred two NRTI are zidovudine and lamivudine

Following tables are common used drug doses and side effects:

Drug	Dose	Side effects
Zidovudine	4 to <9 kg: 12mg/kg PO q 12hr or 8mg/kg PO TID	Anaemia, neutropaenia myopathy, lipoatrophy or lipodystrophy
	≥9kg to 30kg: 9mg/kg PO q 12hr or	Lactic acidosis or severe hepatomegaly with

	6mg/kg TID	steatosis
	≥30kg: 300mg PO q 12hr or 200 mg PO TID	
Lamivudine	Neonates(aged <4 weeks); 2mg/kg PO BID(for prevention of transmission or treatment)	
	Infants 1-3 months: 4mg/kg PO BID	
	3months-16 years: 4mg/kg PO BID or 8mg/kg PO q day, not exceed 300mg/day	
Nevirapine (recommended for < 3 yrs of age)	<15 days : safety and efficacy not established	Hepatotoxicity
	≥15 days to 16years: 150mg/m ² BID; not to exceed 200mg/dose	Severe skin rash and hypersensitivity reaction (Stevens-Johnson syndrome)
	>16 years: as adults; 200mg PO qDay for 2 wk; if no rash or untoward effect occurs,then increase to 200 mg PO BID	
Efavirenz (doses recommendation is only available for ≥ 3 yrs of age)	<3 months: safety and efficacy not established	-Neuropsychiatric adverse effects (bedtimedosing recommended to reduce CNS effects). • Rash (generally mild). • No commercial available liquid. • Limited data on dosing for children aged <3 years. • No data on dosing for children aged <3 months. • Use with caution in adolescent females of childbearing age.
	≥3 months: 3.5kg to <5kg: 100mg PO qDay 5 to <7.5kg : 150mg PO qDay 7.5 to <15kg: 200mg PO qDay 15 to <20kg: 250 mg PO qDay 20 to <25kg: 300mg PO qDay 25 to <32.5kg: 350mg PO qDay 32.5 to <40kg: 400mg PO qDay ≥40kg: 600mg PO qDay	

Lopinavir/ritonavir(Kaletra)	<2 weeks: safety and efficacy not established	Poor palatability of liquid formulation (bitter taste) • Food effect (liquid formulation should be administered with food). • Should not be administered to neonates before a postmenstrual age (first day of the mother's last menstrual period to birth plus the time elapsed after birth) of 42 weeks and a postnatal age ≥ 14 days. • Must be used with caution in patients with preexisting conduction system defects (can prolong PR and QT interval of ECG).
	2weeks-6months: PO solution: 300mg/75mg (LPV/r) per m ² or 16mg/4mg LPV/r per kg PO q 12hr	Electrocardiographic abnormalities (PR and QT interval prolongation, torsades de pointes)
	6months- 18years: 7 to <15kg: 12mg/kg/dose PO q 12hr based on lopinavir component	QT interval prolongation
Lopinavir/ritonavir(Kaletra)	<2 weeks: safety and efficacy not established	Hepatotoxicity
	2weeks-6months: PO solution: 300mg/75mg (LPV/r) per m ² or 16mg/4mg LPV/r per kg PO q 12hr	Pancreatitis
	6months- 18years: 7 to <15kg: 12mg/kg/dose PO q 12hr based on lopinavir component	Risk of prematurity, lipoatrophy or metabolic syndrome, dyslipidaemia or severe diarrhea
	15-<40kg: 10mg/kg/dose q12 hr based on lopinavir component; not exceed 400mg/100mg PO q12hr.	
	>40kg: as adults; 400mg/100mg PO q12hr	
	Oral tablets:6 months-18years: 15-<20kg or 0.6 to 0.8 m ² : 200mg/50mg(2 tab) PO q 12hr	
Cont. Kaletra		

	>20-30kg or 0.8 to 1.2 m ² : 300mg/75mg (3 tab) PO q 12 hr	
	>30-45kg or 1.2 to 1.7m ² : 400mg/100mg PO q12 hr	
	>45kg or >1.7m ² : 400-600mg/100- 150mg PO q12hr	

Counseling:

Once HIV case is suspected paediatric ID speciality involvement is required for pre and post-test counseling

Counseling the parents/patient with HIV is a challenging process that needs to be understood clear as it will have a deep impact in the family and the community as well.

Counseling is needed for the following:

- 1- For any HIV test positive case
 - 2- Pre-testing in high suspicious patients (clinical and LAB findings or social behaviors are supporting)
 - 3- Counseling of HIV mothers after delivery or before if required
- Before starting the counseling one should have the following:

- The room should be easily accessible to target clients.
- Must ensure privacy and confidentiality.
- Must have the HIV test done and confirmed positive more than once/ or HIV test to be done if highly suspicious
- Both parents are available if possible
- HIV nurse and social worker(if required) are available
- The session must be well organized and structured.
- Each session should be flexible in order to meet the requirements of individual clients
- Each session should have a goal-based strategy.
- Any information provided through counseling should be consistent. The counselor therefore needs to have a clear understanding of the facts concerning HIV tests, infection and disease, and maintain close and confidential links with the other health workers who provide care to the client. All counselors should have access to up-to-date information about HIV infection and its management, so that questions from clients can be answered as accurately as possible. Any uncertainties in knowledge should be acknowledged so that new information can be introduced in a consistent framework.

Breaking the news:

- Give results simply and clearly
- Allow time for the result to sink in
- Check that the parents had understood the situation
- Discuss the meaning of the result and its implications
- Explore who to tell and how to tell among the family

- Manage immediate emotional reactions
- Check for immediate follow-up and referral support
- Review options and resources
- Review immediate plans, intentions and actions
 - Providing written information, support services, and a follow-up meeting
 - Hearing about a positive HIV test result can bring out many strong feelings: the client may not be able to concentrate, and may not hear what you are saying. Try to give written information to each person who is able to read, so that he can later read about what he did not hear or understand during the post-test counseling session.
 - No need to ask questions related to risk factors and creat blame environment between couples
 - Assess safety of home for the child : risk of physical &/or Sexual abuse and if there is concern contact the child protection committee in hospital via child life therapist or social worker.
 - Follow-up counseling if required

References

Title of book/ journal/ articles/ Website	Author	Year of publication	Page
http://Aidsinfo.nih.gov/guidelines_pediatrics.pdf		2023	
National manual of HIV in Oman		2014	
Al-Lawati, presentation at the National AIDS Programme Managers meeting, Beirut, 2011		2011	
Guidelines for counselling about HIV infection and disease. WHO AIDS Series 8. WHO, 1990		1990	